# Teaching Machines to Read Signals: AI Meets Ethical Hacking

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# Presentation Overview

- Why Signals Matter In Ethical Hacking
- Traditional Network Layers Vs. RF Signal Layers
- Dataset and AI Model Overview
- Feature Engineering and Feature Selection
- Simulation: Modulation and Encryption Layers
- Key Takeaways

# Why Signals Matter In Ethical Hacking

# What We Learned Vs. Signals Application

**WHAT WE LEARNED**

- Reconnaissance (PASSIVE :D)
- Enumeration
- Gaining Access
- For servers, databases, and web applications

# What We Learned
# Vs.
# Signals Application

**SIGNALS**

- Information travels over wavelengths (Signals)
- Signals carry encoded information (Bitstream)
- Transmitter alters bitstream
- If Receiver knows altercations, bitstream is received and information is understandable

# What We Learned
# Vs.
# Signals Application

**THE BRIDGE**

- We practiced with information that arrived or already existed within a machine
- Signals allows us to see information during transit, but can be protected in layers like what we learned

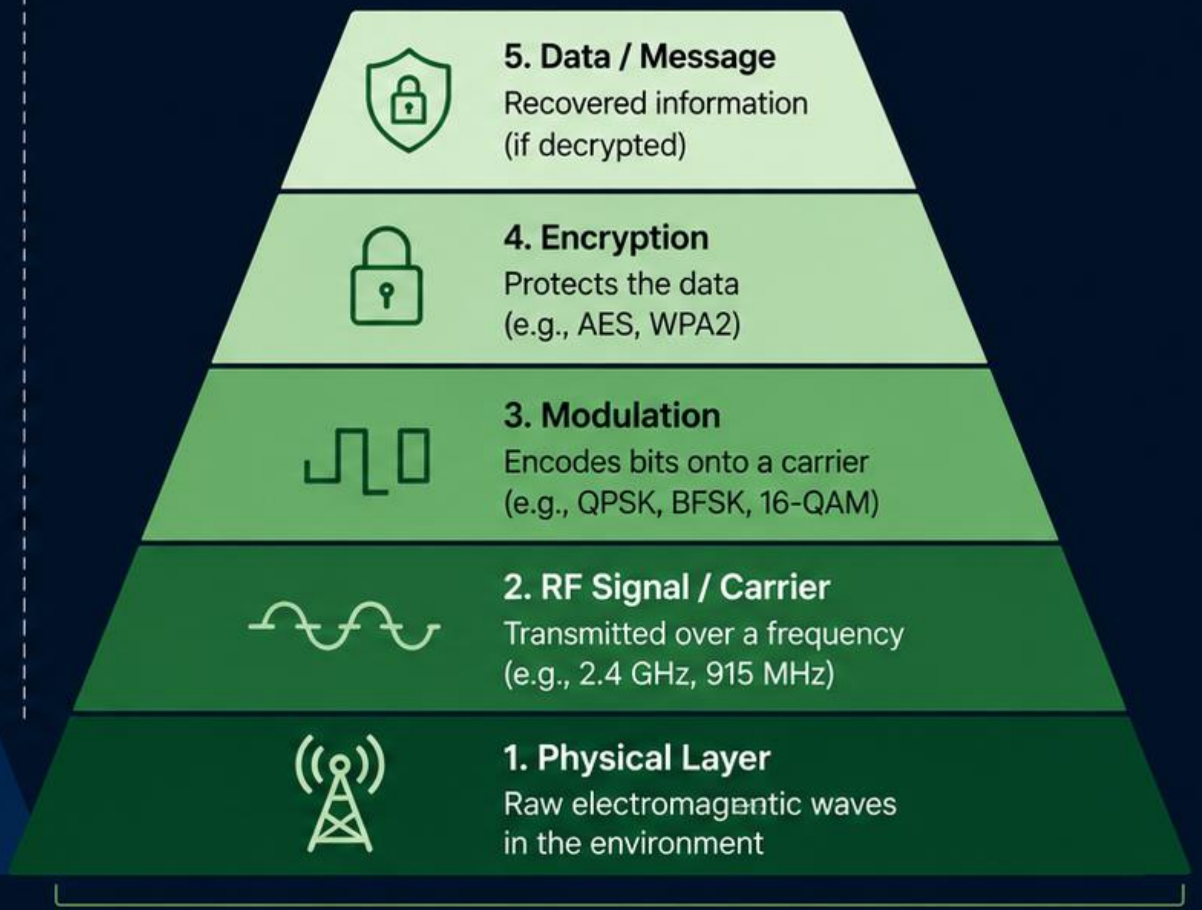
# Traditional Network Layers
# Vs.
# RF Signal Layers

# Traditional Network Layers vs. RF Signal Layers

## 5 Ethical Hacking Layers


Layers 1–3 are commonly covered in class.
Layers 4–5 are beyond the current scope.

## RF Signal Layers (from Physical to Data)



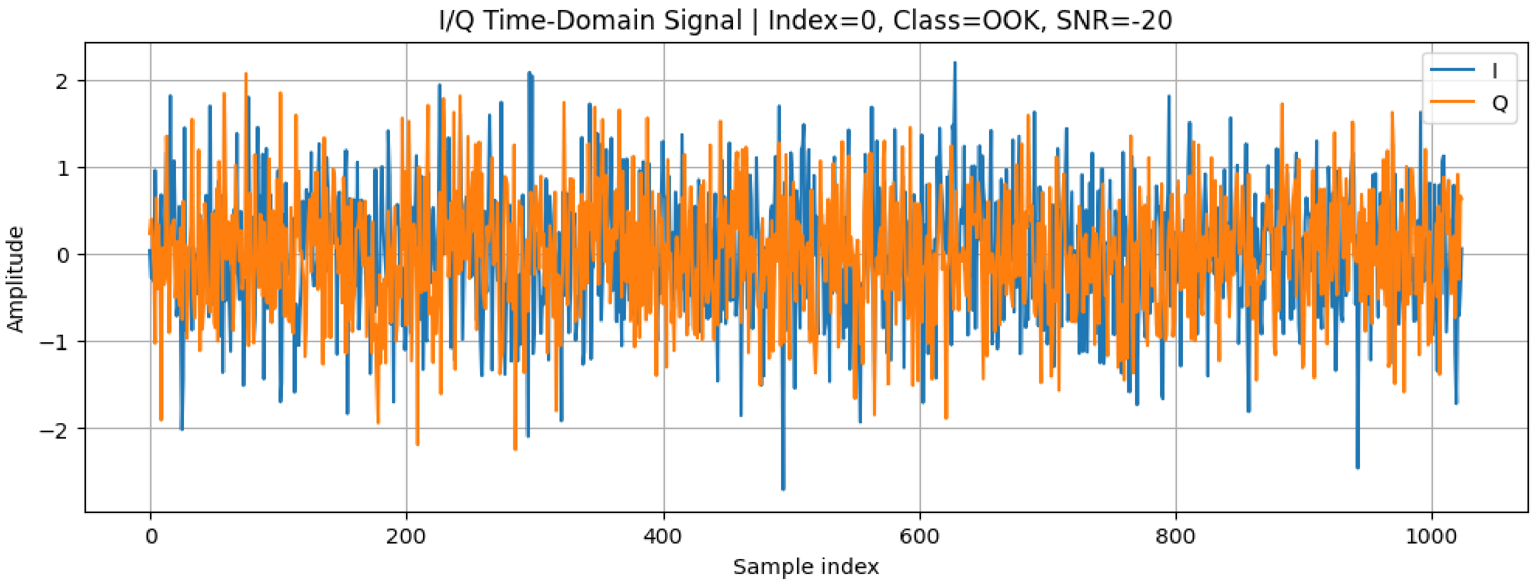

Each higher layer depends on the correct interpretation of the layers **below** it.

Just as ethical hacking follows a layered approach in networks and systems,
RF analysis follows layered signal processing from the physical signal up to meaningful data.

# Dataset and AI Model Overview

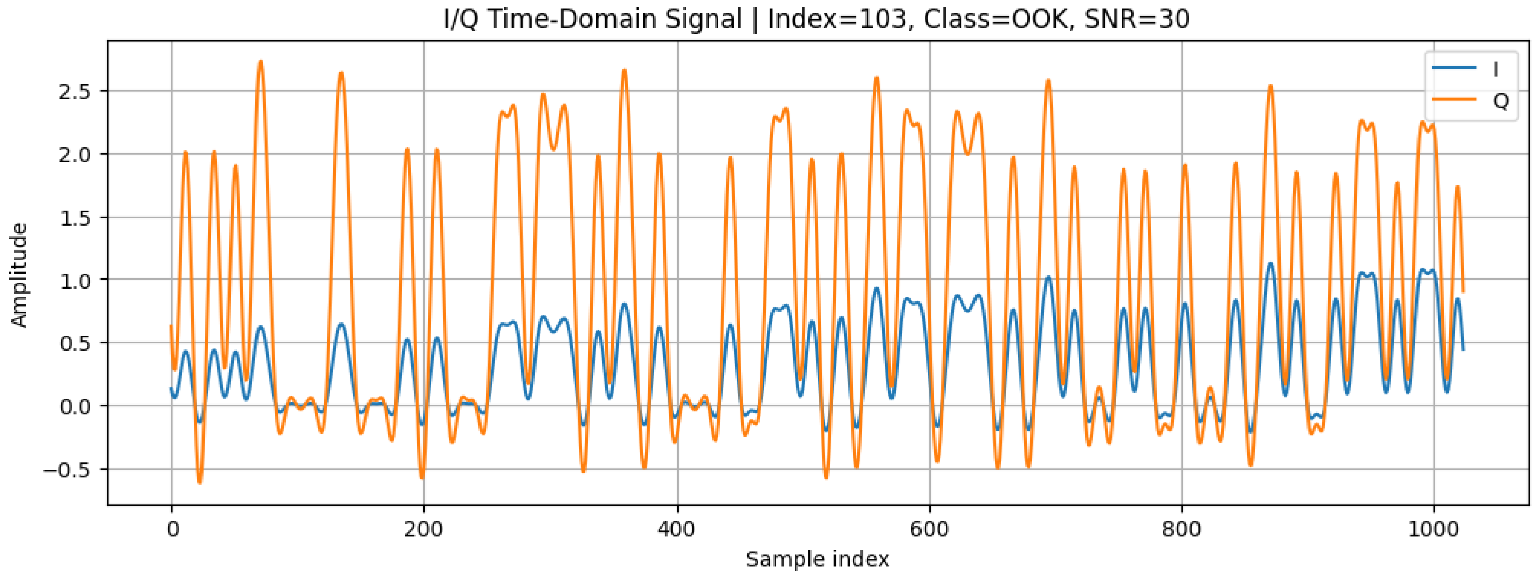
# Full Dataset

- [DeepSig's RADIOML 2018.01A]()
- 2,555,904 Signals
- 24 Modulation Types
  - Uniform Distribution
- 3 Features
  - I/Q (1024 array * 2)
  - Modulation ID
  - Signal-to-Noise Ratio (SNR: [-20, 30])



# Reduced Dataset

- DeepSig's RADIOML 2018.01A
- 24,000 Signals
- 24 Modulation Types
  - Uniform Distribution
- 3 Features
  - I/Q (1024 array * 2)
  - Modulation ID
  - Signal-to-Noise Ratio (SNR=30)



# Random Forest Classifier

An ensemble of decision trees that improves accuracy and reduces overfitting by averaging the predictions of multiple trees.

### 1 Bootstrap Sampling

Randomly sample (with replacement) from the training data to create multiple datasets.

### 2 Train Multiple Trees

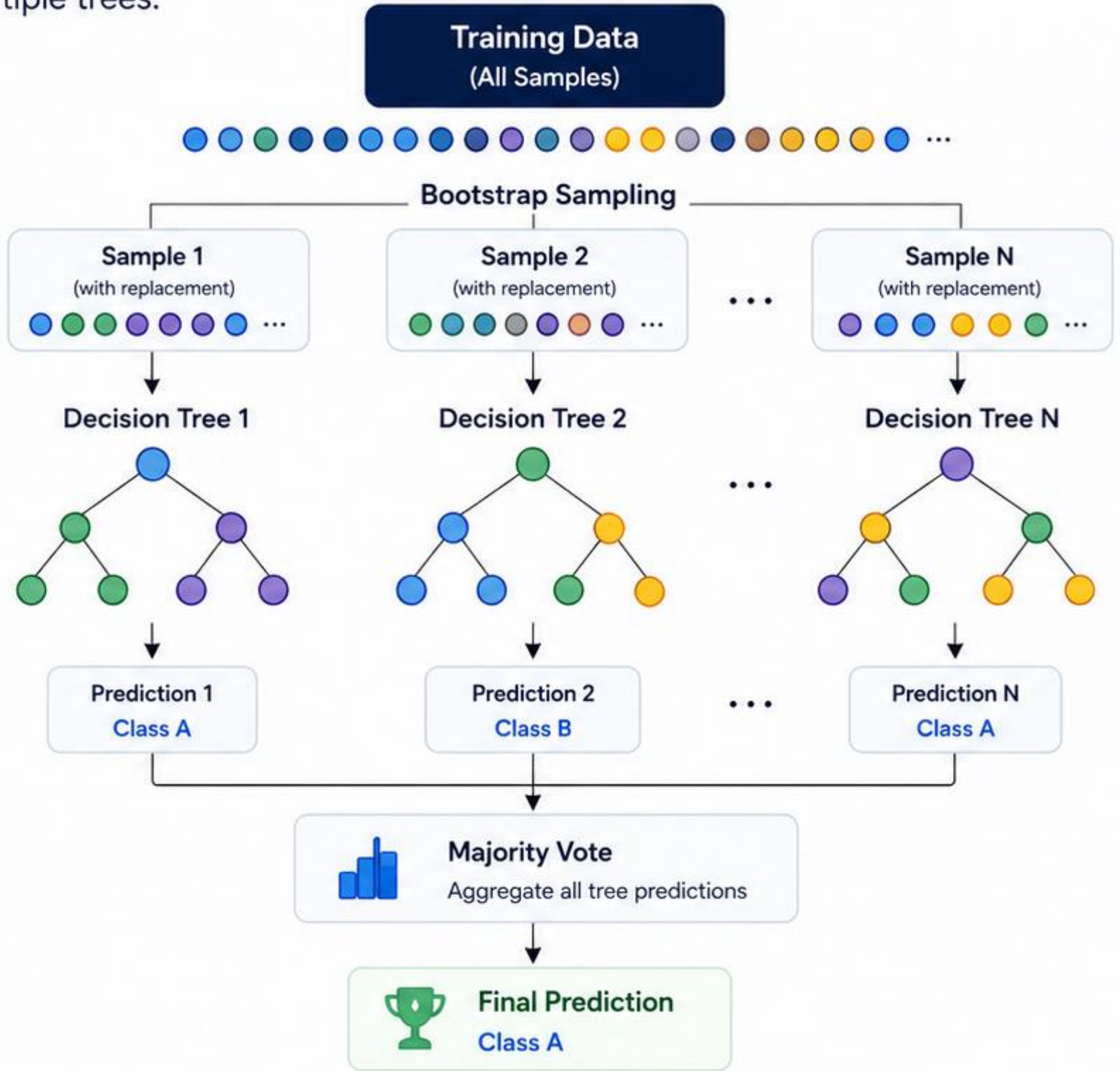
Train a decision tree on each bootstrap sample. Each tree sees a different subset of the data.

### 3 Aggregate Predictions

Each tree makes a prediction. The final output is determined by majority vote (classification).

### 4 Final Prediction

The class with the most votes is the Random Forest prediction.




## Why Random Forest?

**High Accuracy**
Combines many trees to reduce error.

**Robust to Overfitting**
Averaging reduces variance compared to a single tree.

**Handles Large Datasets**
Works well with high-dimensional data and missing values.

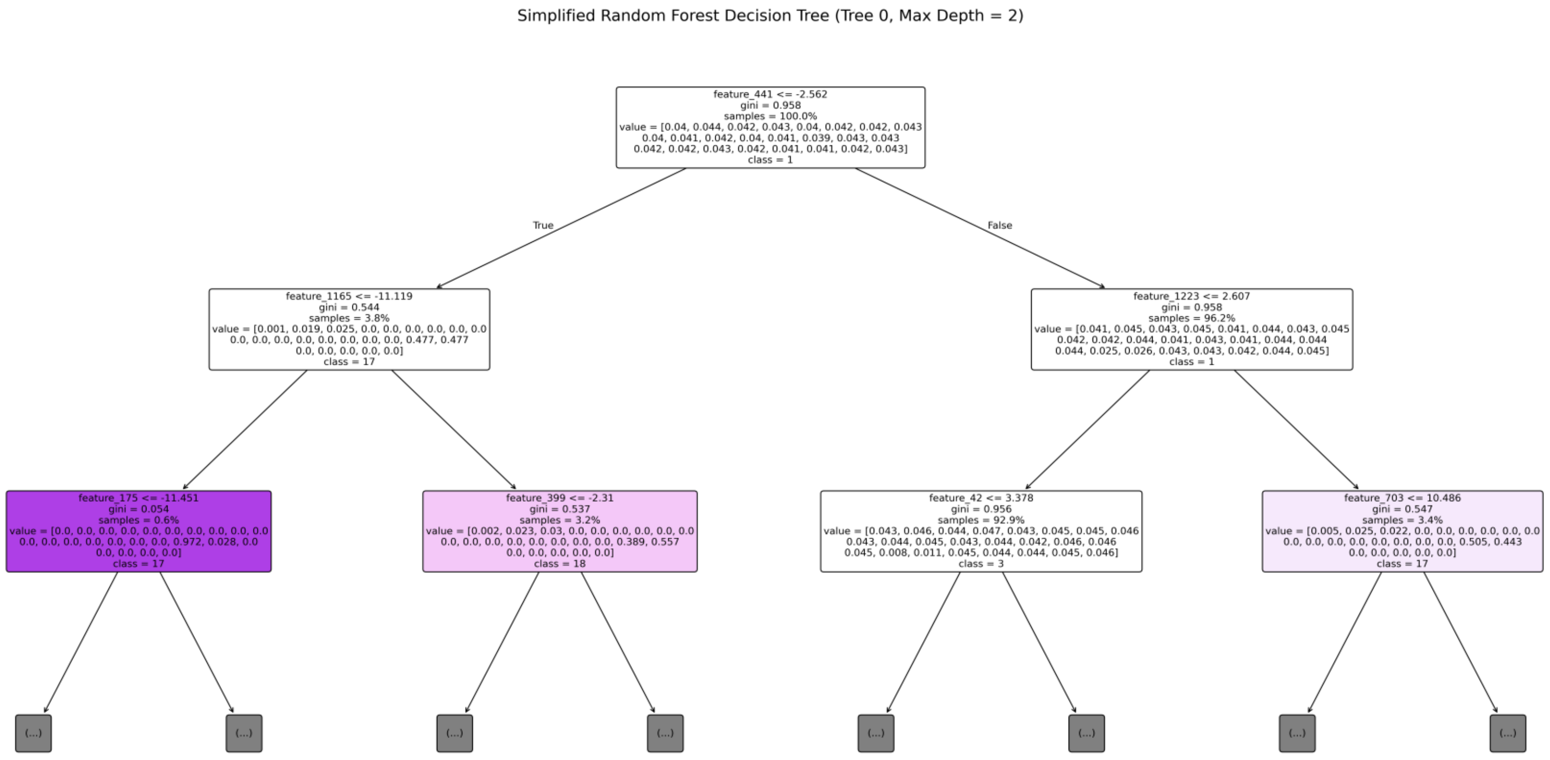
**Feature Importance**
Provides insight into which features matter most.

## Key Points

- Uses bagging (bootstrap aggregating)
- Each tree is trained on a random subset
- Predictions are combined by majority vote
- Reduces overfitting and improves generalization

# Baseline Model

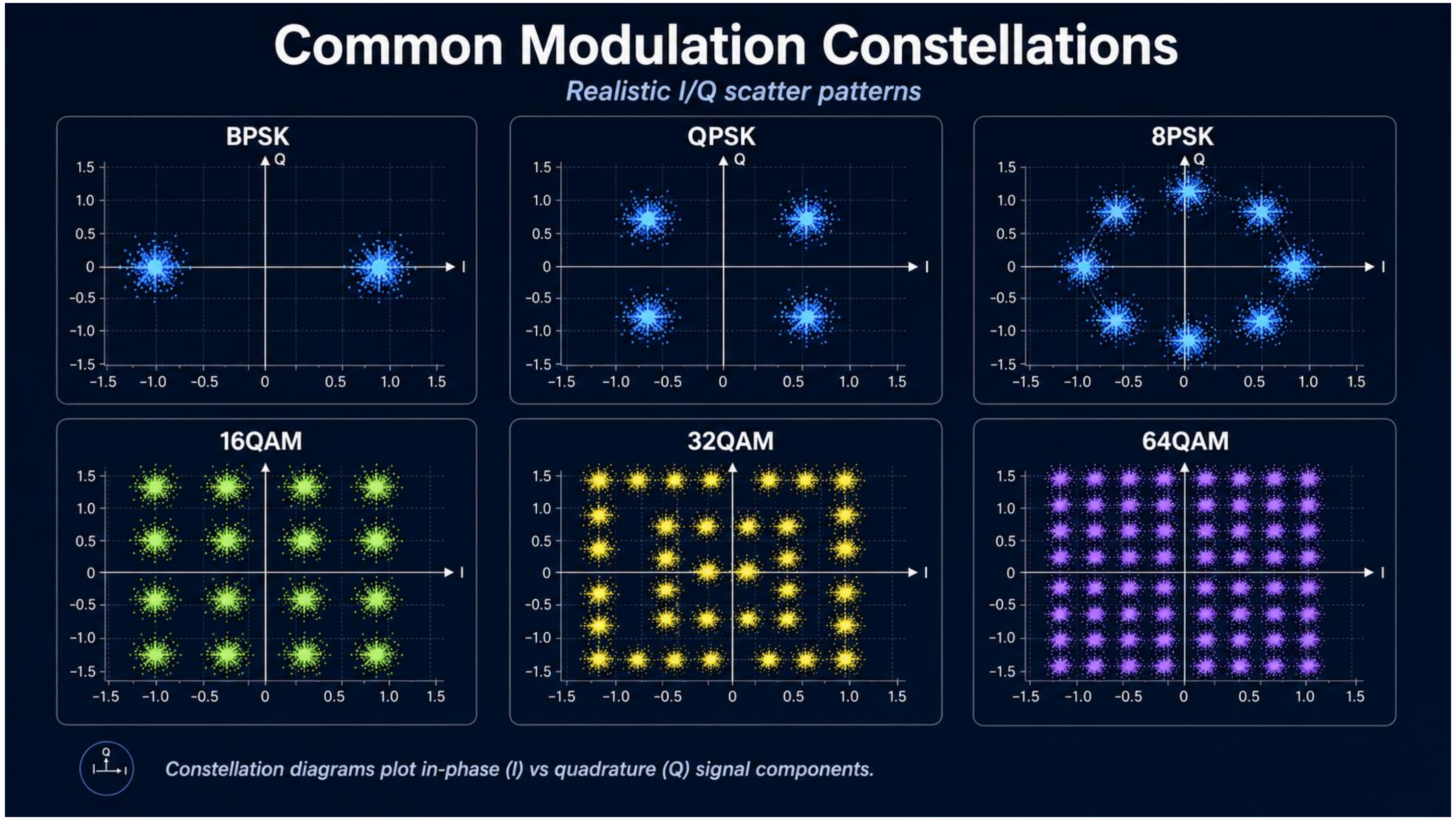
- Input:
  - 2,048 Features
  - 24,000 Signals
  - 24 Modulation Types
    - Uniform Distribution
- Output:
  - 1of24
    Modulation Types
- Performance:
  - Accuracy ~36.21%



# Feature Engineering and Feature Selection

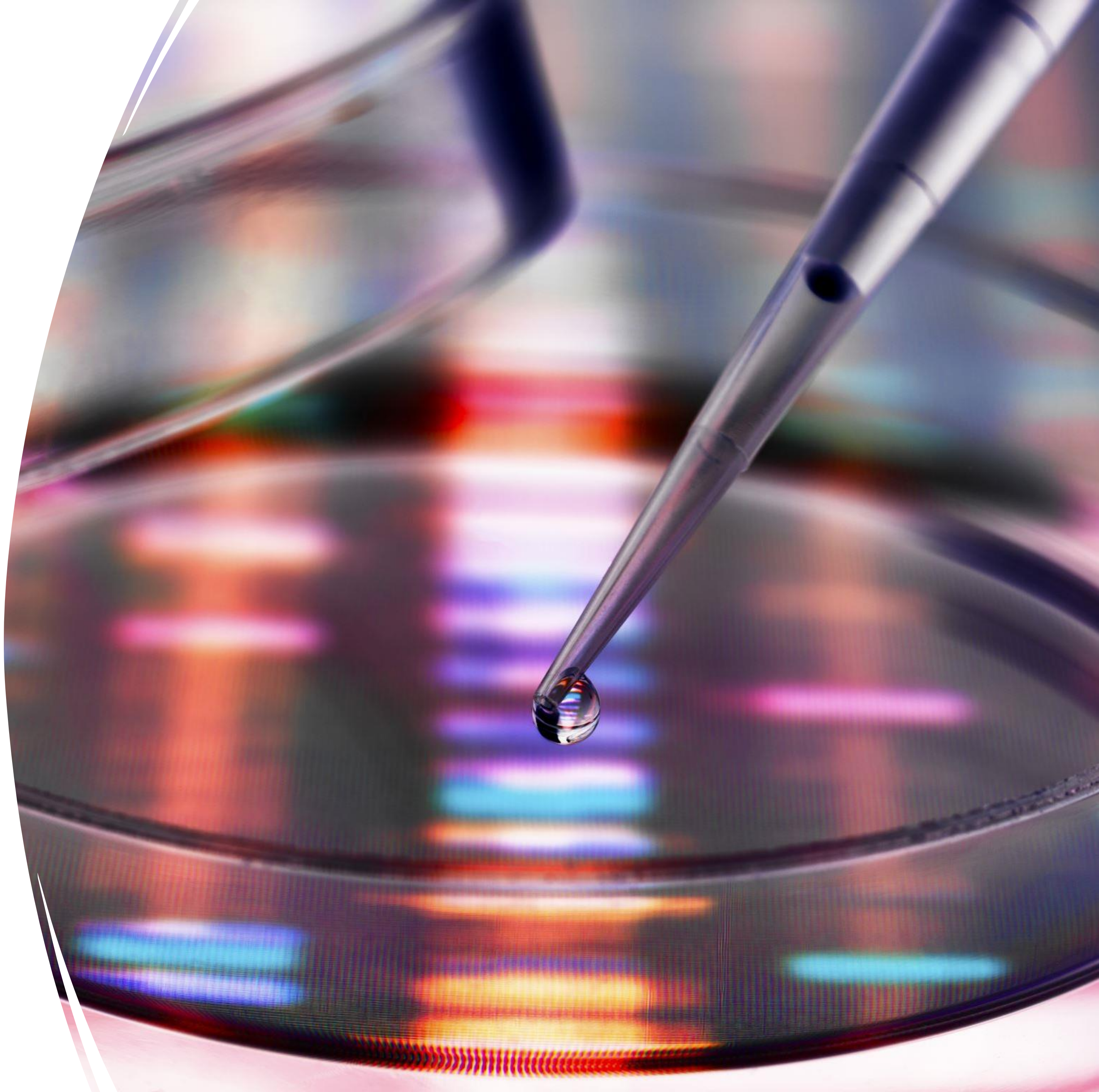
# The SIGINT Pipeline

- SIGINTers apply complex mathematics to garner more information about signals like:
  - I/Q stats
  - Magnitude/Power
  - Phase/Frequency
  - FFT/Spectral Features
  - Constellations



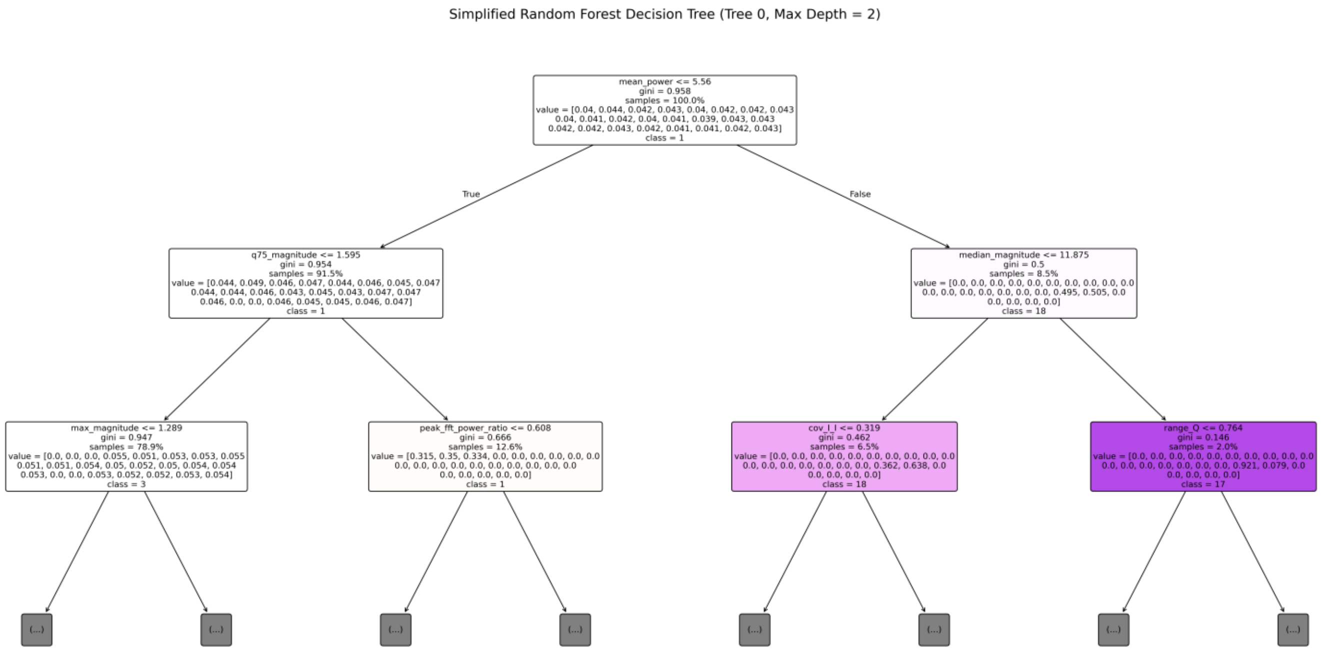
# Feature Engineered Dataset

- 24,000 Signals
- 24 Modulation Types
  - Uniform Distribution
- 105 Features
  - I/Q stats
  - Magnitude/Power
  - Phase/Frequency
  - FFT/Spectral Features
  - Constellations



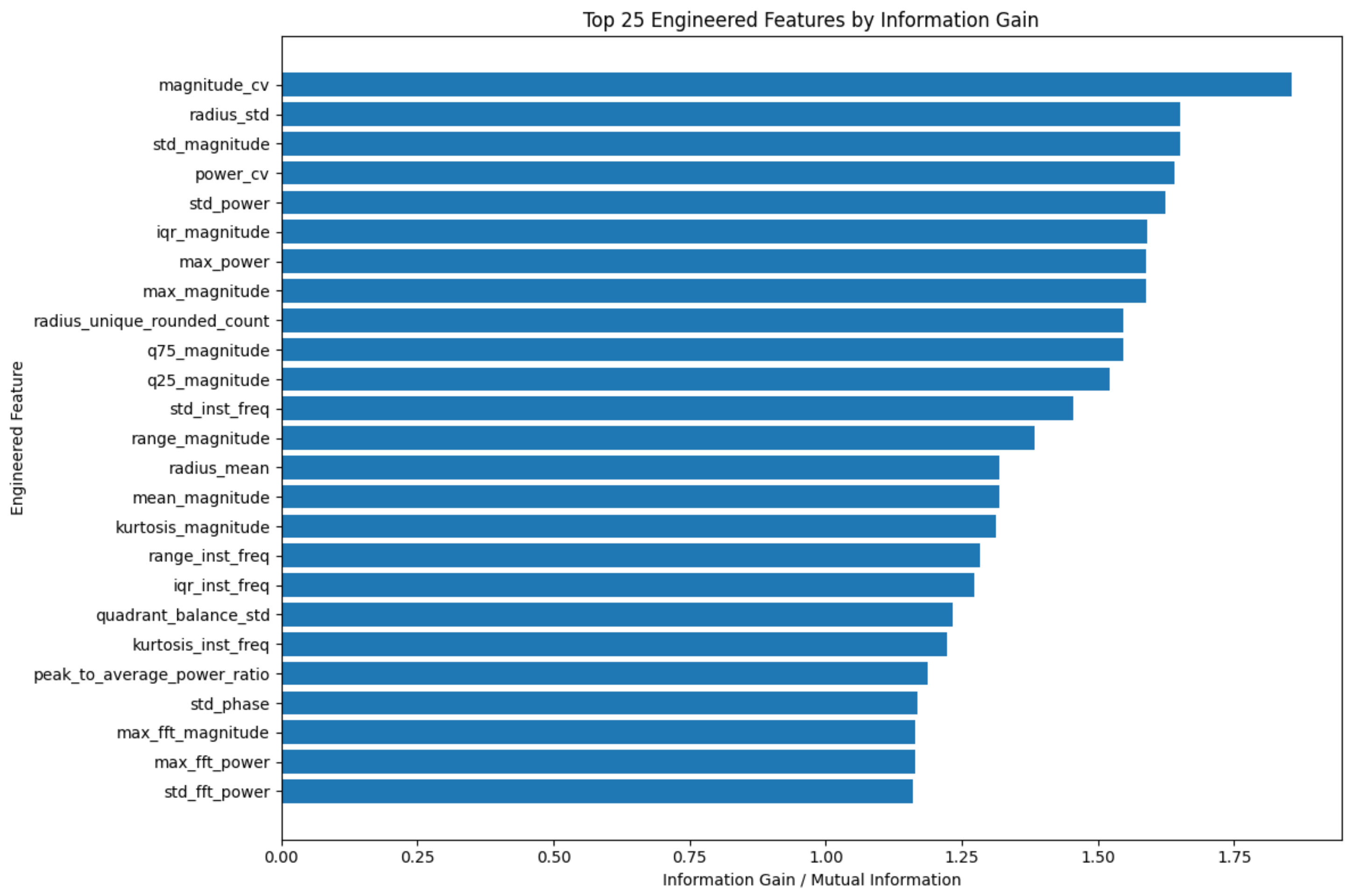
# Feature Engineered Model

- Input:
  - 105 Features
  - 24,000 Signals
  - 24 Modulation Types
    - Uniform Distribution
- Output:
  - 1of24 Modulation Types
- Performance:
  - Accuracy ~72.17%



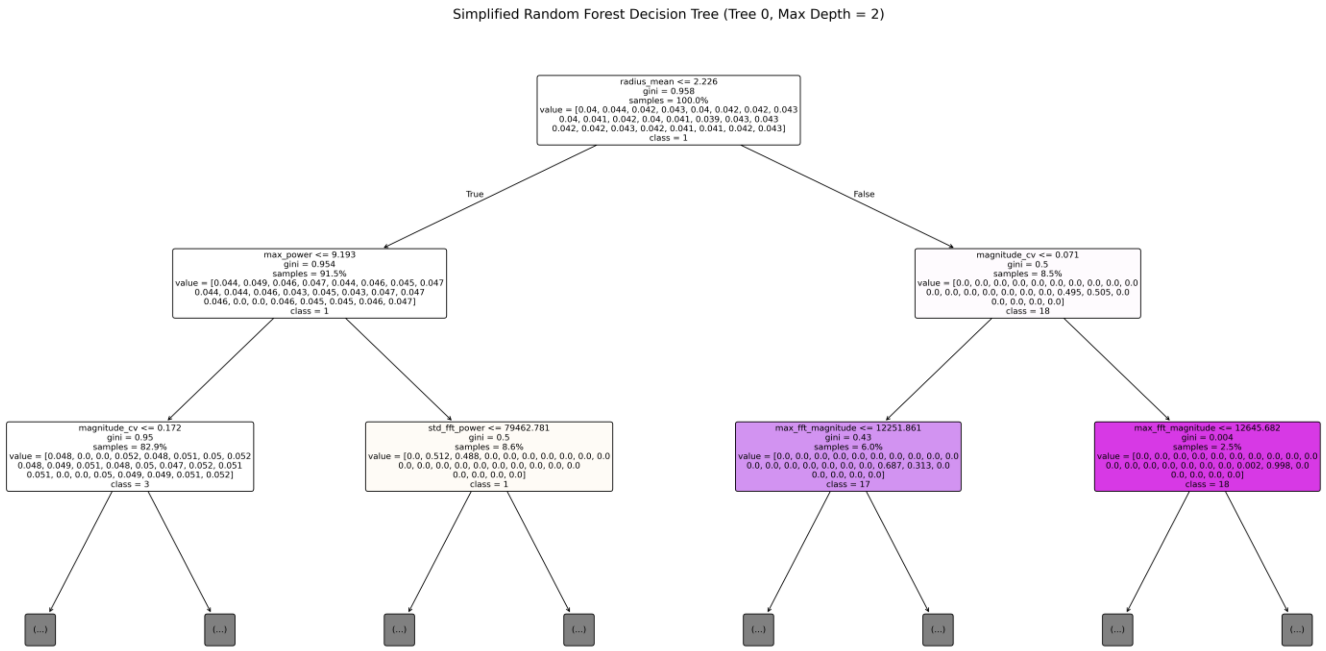
# Feature Selection

- Each feature can generate an information gain score
- Higher information gain leads to higher confidence in decisions



# Top 25 Feature Model

- Input:
  - 25 Features
  - 24,000 Signals
  - 24 Modulation Types
    - Uniform Distribution
- Output:
  - 1of24
    Modulation Types
- Performance:
  - Accuracy ~69.37%



# Simulation: Modulation and Encryption Layers

# Wrong Mod/Encryption

# Correct Mod/Encryption

| correct_modulation | correct_encryption | successful_recovery | original_bit_error_rate | recovered_text |
|---:|---:|---:|---:|---:|
| True | True | True | 0.000000 | HELLO RF WORLD |
| False | True | False | 0.473214 | ა��\t��ecretse |
| True | False | False | 0.312500 | LR@PM JF:RXIQF |
| False | False | False | 0.446429 | گّ�� ��eywrong |

# Key Takeaways

- Traditional Ethical Hacking and RF Signal Hacking are almost identical in practice
- RFC shows potential to classifying modulation type, but needs more training (Particularly with noisy data)
- Higher quality matters more than higher quantity data

# Resources Used

- DeepSig's RADIOML 2018.01A (Original Dataset)
- PySDR (General SDR Knowledge)
- Various Sources on SDR (Supplementary SDR Knowledge, in writeup)
- Microsoft Powerpoint AI Designer (Slide Setup)
- ChatGPT Image Creation (Slide Setup)


Thank You!